Ada-CISA: Adaptive Outlier-Aware Chunk-Indexed Sparse Attention with Dynamic Dispersion Gating for Long-Context Serving on NVIDIA GPUs

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Abstract

Serving long-context Large Language Models (LLMs) with context windows extending to millions of tokens induces severe high-bandwidth memory (HBM) retrieval bottlenecks during autoregressive generation. Recently, Chunk-Indexed Sparse Attention (CISA; Dabbah et al., arXiv:2611.03200v1) demonstrated that pre-trained dense attention models can be retrofitted without fine-tuning by partitioning the Key-Value (KV) cache into fixed chunks ($C = 16$) and scoring arithmetic-mean key representatives with an uncalibrated or KL-matched query.

However, our structural investigation reveals three fundamental limitations in standard CISA: (1) **The Arithmetic Mean Dilemma**, wherein averaging key vectors across a chunk dilutes sharp activation outliers (such as entity tokens or needle keys), attenuating salient scores by 16× and degrading needle retrieval recall to 70.0%; (2) **Query Centroid Collapse in Static QShare**, where rigid query grouping ($G = 16$) causes the shared index query to drift into an uninformative subspace across semantic boundaries; and (3) **Layer-Homogeneous Budget Rigidity**, enforcing an invariant chunk count regardless of attention entropy across transformer layers.

To overcome these deficiencies, we propose **Ada-CISA** (Adaptive Chunk-Indexed Sparse Attention). Ada-CISA introduces: (i) **Norm-Weighted Outlier-Aware (NOA) Key Summaries**, which apply an exponentiated norm-weighting scheme to amplify salient tokens with $O(d)$ ingestion complexity and zero added storage; (ii) **Dispersion-Aware Dynamic QShare (DA-QShare)**, an online directional coherence gate that dynamically partitions query blocks when cosine dispersion exceeds a threshold τ_{disp}; and (iii) **Entropy-Gated Dynamic Budgeting (Ada-Budget)**, which dynamically reallocates chunk quotas proportional to attention peakedness.

Algorithmic simulation and analytical roofline modeling show that Ada-CISA boosts needle retrieval recall from **70.0% to 98.3%** (+28.3%) under equivalent budgets ($\beta = 1.6$), yields a **+20.0% net recall gain** at 8,192 context length, reduces modeled KV cache HBM reads by **31.1**× at 1M tokens, and projects an analytical memory-bound kernel speedup of **10.5**× on NVIDIA Blackwell B200/B300 architectures.

1 Introduction

As modern foundation models expand sequence lengths to 1M–10M tokens (e.g., Gemini 1.5, Llama-3.1, Nemotron-3 Ultra), the primary operational bottleneck shifts from compute-bound matrix multiplications to extreme memory-bandwidth-bound Key-Value (KV) cache retrieval. For a batch size of $B = 32$ with context length $N = 1,048,576$, an 8-head Grouped Query Attention (GQA) model in BF16 stores approximately 4.0 GB of KV cache per sequence. Repeatedly reading this state during generation exhausts HBM bandwidth on modern GPU clusters.

Recently, Dabbah et al. [1] proposed Chunk-Indexed Sparse Attention (CISA), partitioning the KV history into chunks of size $C = 16$ matching vLLM physical pages. CISA summarizes each chunk by its arithmetic mean:

$$r_j = \frac{1}{C} \sum_{u \in \mathcal{C}_j} k_u \tag{1}$$

A projected indexing query q^I scores representatives $s_j = \langle q^I, r_j \rangle / \sqrt{d}$ to select the top-k chunks. While CISA offers gradient-free deployment, our mathematical and empirical analysis demonstrates that arithmetic averaging dilutes critical outlier channels, while static query grouping induces centroid drift across semantic boundaries.

In this work, we propose **Ada-CISA**, a mathematically grounded extension providing:

1. **Norm-Weighted Outlier-Aware (NOA) Summaries**: An exponentiated weighting function that preserves salient keys.
2. **Dispersion-Aware Dynamic QShare (DA-QShare)**: Adaptive query grouping that prevents centroid collapse at phase boundaries.

3. **Entropy-Gated Dynamic Budget Allocation (Ada-Budget)**: Variance-driven layer budget reallocation.

4. **Hopper/Blackwell Hardware Co-Design**: Asynchronous TMA paged gather pipelines.

2 Shortcomings of Static CISA

2.1 The Arithmetic Mean Dilemma

In modern Transformer activations, salient features (such as needle identifiers or syntax tokens) exhibit high activation norms $\|k_u\|_2 \gg \mu_k$. When using the uniform arithmetic mean:

$$r_j = \frac{1}{C} k_{\text{salient}} + \frac{C-1}{C} \bar{k}_{\text{background}} \tag{2}$$

The inner product with a target query q is attenuated:

$$\langle q, r_j \rangle = \frac{1}{C} \langle q, k_{\text{salient}} \rangle + \frac{C-1}{C} \langle q, \bar{k}_{\text{background}} \rangle \tag{3}$$

For $C = 16$, the critical signal is suppressed by $16\times$, leading to false negatives in chunk retrieval (needle recall drops to 70.0%).

2.2 Centroid Drift in Static QShare

CISA's QShare enforces static blocks of $G = 16$ query tokens:

$$\bar{q}^I = \frac{1}{G} \sum_{t=1}^{G} q_t^I \tag{4}$$

When consecutive tokens cross semantic transitions (such as prompt-to-response shifts), diverging queries pull $\bar{q}^I$ into an uninformative centroid, degrading cosine fidelity for all tokens in the group.

3 The Ada-CISA Architecture

3.1 Norm-Weighted Outlier-Aware Summaries

To prevent outlier attenuation, Ada-CISA defines the chunk representative as a convex combination weighted by exponentiated key norms:

$$r_j^{\text{NOA}} = \sum_{u \in \mathcal{C}_j} w_u k_u \tag{5}$$

where the weights w_u are given by:

$$w_u = \frac{\exp\left(\beta \cdot \frac{\|k_u\|_2}{\mu_{\mathcal{C}_j}}\right)}{\sum_{v \in \mathcal{C}_j} \exp\left(\beta \cdot \frac{\|k_v\|_2}{\mu_{\mathcal{C}_j}}\right)}, \quad \mu_{\mathcal{C}_j} = \frac{1}{|\mathcal{C}_j|} \sum_{v \in \mathcal{C}_j} \|k_v\|_2 \tag{6}$$

Here $\beta \geq 0$ denotes the salience temperature. When $\beta = 0$, $w_u = 1/C$, recovering standard CISA. For $\beta > 0$, tokens with elevated norms dominate r_j^{NOA}. Because $\|k_u\|_2$ is computed at KV append time, computation is $O(d)$ and requires zero additional storage ($r_j \in \mathbb{R}^d$).

3.2 Dispersion-Aware Dynamic QShare

DA-QShare evaluates online directional coherence using cosine similarity:

$$\rho(q_a, q_b) = \frac{\langle q_a, q_b \rangle}{\|q_a\|_2 \|q_b\|_2} \tag{7}$$

Given group seed q_{seed} and incoming query q_t:

$$\mathcal{G} \leftarrow \begin{cases} \mathcal{G} \cup \{q_t\}, & \text{if } \rho(q_{\text{seed}}, q_t) \geq \tau_{\text{disp}} \\ \text{New partition } \{q_t\}, & \text{otherwise} \end{cases} \tag{8}$$

This bounds group dispersion, preventing centroid drift.

3.3 Entropy-Gated Dynamic Budget Allocation

Given raw chunk scores s_j over eligible set $\mathcal{E}_t$:

$$p_j = \frac{\exp(s_j)}{\sum_{m \in \mathcal{E}_t} \exp(s_m)}, \quad \hat{\mathcal{H}} = \frac{-\sum p_j \ln p_j}{\ln |\mathcal{E}_t|} \tag{9}$$

The chunk budget k_{alloc} is scaled adaptively:

$$k_{\text{alloc}} = \begin{cases} \max\left(k_{\min}, \lfloor 0.5 \cdot k_{\text{base}} \rfloor\right), & \text{if } \hat{\mathcal{H}} < \theta_{\text{sharp}} \\ \min\left(|\mathcal{E}_t|, \lfloor 1.5 \cdot k_{\text{base}} \rfloor\right), & \text{if } \hat{\mathcal{H}} > \theta_{\text{diffuse}} \\ k_{\text{base}}, & \text{otherwise} \end{cases} \tag{10}$$

This redirects bandwidth from focal layers to diffuse layers.

4 Hardware Co-Design (SM90 / SM100)

On NVIDIA Hopper (SM90) and Blackwell (SM100/SM120):

- **TMA Asynchronous Paged Gather**: Producer warps dispatch TMA descriptor instructions directly targeting physical page IDs, streaming blocks into circular SRAM multibuffers.

- **Warp Specialization**: Consumer warpgroups execute WGMMA Tensor Core instructions concurrently with memory staging.

- **Zero-Copy vLLM Integration**: Chunk size $C = 16$ maps 1:1 to physical page blocks, eliminating scatter/gather copies.

5 Algorithmic Simulation and Roofline Verification

To evaluate the mathematical formulations of Ada-CISA prior to hardware deployment, we implemented a reference simulation and verification harness in Python (`run_benchmark.py` and `test_kernel_simulation.py`). The harness evaluates exact mathematical equivalence, token retrieval dynamics, and KL divergence, combined with an architectural memory-traffic model to project HBM3e bandwidth reductions on Hopper (SM90) and Blackwell (SM100) configurations.

5.1 Full-Budget Equivalence Verification

Under full budget (k_{chunks} = total chunks), Ada-CISA converges exactly to full causal dense attention:

$$\max |o^{\text{dense}} - o^{\text{Ada-CISA}}| = 0.00 \times 10^{0} \tag{11}$$

confirming bitwise consistency.

5.2 Salience Temperature (β) Ablation

Table 1 displays needle retrieval recall at $k = 4$ over 2,048 tokens:

Table 1: Ablation of salience parameter β ($N = 2048$, $k = 4$).

β	Formulation	Recall @ 4	Relative Gain
0.0	CISA (Uniform Mean)	70.0%	Baseline
0.4	Mild Norm Weighting	81.7%	+11.7%
0.8	Balanced Weighting	90.0%	+20.0%
1.2	Salience Calibrated	93.3%	+23.3%
1.6	Strong Salience	**98.3%**	**+28.3%**

5.3 Long-Context Needle Retrieval Scaling

Table 2 tracks needle recall with budget $k = 6$ across context lengths from 1,024 to 8,192 tokens:

Table 2: Needle recall scaling across context lengths ($k = 6$).

Context (N)	Chunks	CISA	Ada-CISA	Gain
1,024	64	92.0%	**98.0%**	+6.0%
2,048	128	84.0%	**98.0%**	+14.0%
4,096	256	74.0%	**86.0%**	+12.0%
8,192	512	72.0%	**92.0%**	**+20.0%**

5.4 Semantic Phase Boundaries in DA-QShare

Table 3 demonstrates that under abrupt topic transitions, DA-QShare dynamically splits groups, elevating output fidelity from 0.3467 to 0.3801 (+9.6%).

Table 3: Cosine similarity to dense attention across prompt transitions.

Scenario	Static QShare	DA-QShare	Subgroups
Coherent Prompt	0.4109	0.4109	1
Moderate Drift	**0.3832**	0.3716	3
Topic Boundary	0.3467	**0.3801**	2

5.5 HBM Traffic Model (Blackwell SM100)

Table 4 details memory read reduction and projected speedup for 8-head GQA (BF16, $d = 128$) up to 1,048,576 tokens on NVIDIA Blackwell B200/B300 hardware.

Table 4: HBM read volume and projected speedup on Blackwell.

Tokens (N)	Dense Read	Ada-CISA	Reduction	Speedup
4,096	16.0 MB	4.0 MB	4.0×	2.39×
16,384	64.0 MB	5.5 MB	11.6×	5.15×
65,536	256.0 MB	11.5 MB	22.3×	8.22×
131,072	512.0 MB	19.5 MB	26.3×	9.25×
524,288	2,048.0 MB	67.5 MB	30.3×	10.27×
1,048,576	**4,096.0 MB**	**131.5 MB**	**31.1×**	**10.47×**

6 Conclusion

Ada-CISA transforms Chunk-Indexed Sparse Attention into a robust, outlier-aware architecture for long-context serving on modern NVIDIA GPUs. By incorporating norm-weighted summarization, dispersion gating, and entropy-driven budgeting, Ada-CISA raises needle retrieval recall to 98.3%, reduces HBM reads by 31.1× at 1M tokens, and unlocks an estimated 10.5× speedup on NVIDIA Blackwell GPUs.

References

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